

Panthaweeekan Somngam

Software Engineer

Bangkok, Thailand (GMT+7) | Open to Remote / Hybrid

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EN (Intermediate) | TH (Native)

PROFESSIONAL EXPERIENCE

Associate Software Engineer

May 2025 – Present

SCB TechX, Bangkok, Thailand (Hybrid)

- Developing a Go microservice for token validation, authentication, and request routing behind Kong, as part of a phased migration from a legacy on-premise gateway toward SCB Easy's 10M+ user platform.
- Designed the plugin system, routing engine, and observability infrastructure (Prometheus metrics, structured logging) for the new gateway, currently progressing through phased migration to production.
- Designed and developed an enterprise asset management platform with multi-stage approval workflows, role-based access control for six user roles, financial reporting dashboards, and real-time notifications.
- Leveraged AI-assisted development tools across code generation, unit-testing, refactoring, architecture review, and documentation to accelerate delivery on complex features.

Full Stack Developer (Work-Study Scholarship)

2023 – 2025

Chiang Mai University, Faculty of Engineering, Chiang Mai, Thailand

- Designed and deployed two production web applications used across the faculty, owning the full lifecycle from stakeholder requirements through containerized deployment.
- Developed a real-time room reservation system with role-based access, an admin dashboard, and mobile push notifications — adopted by the faculty for daily scheduling and still in active use today.
- Developed a multi-stage certificate approval workflow that digitized a previously paper-based process, streamlining manual approval turnaround.
- Applied hexagonal architecture across all projects, building RESTful APIs in Go with React/TypeScript frontends and PostgreSQL.

PROJECTS

API Gateway Microservice — SCB TechX

2025 – Present

Go, Fiber, PostgreSQL, Redis, Kong, Prometheus, Kubernetes

- Built a token service and routing layer in Go behind Kong to incrementally replace a legacy gateway, implementing token authentication and plugin-based request transformation.
- Designed and prototyped a declarative config-sync approach (Kong deckK pattern: YAML route definitions with diff and rollback) and a per-service circuit-breaking strategy, targeting GitOps-style config management and safe phased rollout across the migration's 30+ services.
- Structured the service using Hexagonal Architecture with strict layer separation, dependency injection, and interface-driven design — keeping core business logic fully testable and decoupled from Kong, Redis, and all infrastructure dependencies.

Inventory Asset Management System — SCB TechX

2025

Ruby on Rails, React, TypeScript, PostgreSQL, Azure AD, Docker

- Built a database-driven workflow engine to handle complex request lifecycles across four parallel workflows (request, spare, repair, return), enforcing approval chains per role and step.
- Implemented role-based access control for six user roles with Azure AD single sign-on.
- Built reporting dashboards aggregating asset requests, utilization, financial summaries, and processing efficiency, with Excel and CSV export.
- Developed the frontend with reusable components, partially generated type-safe API clients, and an installable PWA with push notifications.

SD-Booking — Room Reservation System — Chiang Mai University

2023 – 2024

Go, React, TypeScript, PostgreSQL, PWA, Docker

- Built a real-time room booking platform used across the faculty, with role-based access, booking conflict detection, and mobile push notifications — adopted for daily operations and still in use today.

FitJourney — Fitness Tracking Platform — Side Project

2026

Next.js, Supabase, Tailwind CSS, OpenAI API, Vercel

- Shipped a full-stack fitness PWA addressing a personal gap in daily tracking — consolidating workout logging, nutrition, body metrics, and AI-powered meal scanning into a single self-hosted tool.
- Designed analytics dashboards for performance trends and calorie tracking; architected as a monorepo with Supabase (auth, DB, storage) and Vercel edge deployment for fast cold starts.

Twitch Chat TTS Bot — Side Project

2026

Bun, Microsoft Edge TTS, tmi.js, FFplay

- Engineered a real-time Twitch TTS bot that pipes synthesized audio directly from memory to playback, eliminating disk I/O and reducing per-message overhead versus file-based approaches — supporting Thai and English neural voices via Microsoft Edge TTS.
- Built a message priority queue with anti-spam filtering and automated OAuth token refresh to handle high-volume chat reliably without manual intervention.

EDUCATION

Bachelor of Engineering in Computer Engineering

Graduated 2025

Chiang Mai University

Second-Class Honours GPA: 3.32/4.00

Thesis: *LongPlan* — Study plan validator using graph algorithms (topological sort, cycle detection) and curriculum rule enforcement. Built in Go with React frontend.

SKILLS

Languages: Go, TypeScript, JavaScript, Ruby, Python, SQL, HTML/CSS

Familiar: Java, Haskell

Frameworks: Fiber, Gin, Ruby on Rails, React, Next.js, Node.js, Vite, Tailwind CSS, MUI, TanStack Query

Infrastructure: PostgreSQL, Redis, Docker, Kubernetes, AWS, Kong API Gateway, CA Layer 7, Nginx, Supabase, Vercel, Linux

Observability: Prometheus, Grafana, ELK Stack

Tools: Git, GitHub Actions, Jenkins, HashiCorp Vault

Practices: System Design, Data Structures & Algorithms, REST APIs, Microservices, CI/CD, TDD, SOLID, OOP, Agile/Scrum